

SHETH L.U.J & M V COLLEGE

PRACTICAL NO 6 MOD 2

The screenshot displays the RStudio interface with a script editor, console, and environment pane. The script performs a paired t-test on the 'retail_store_sales.csv' dataset, comparing 'Total Spent' and 'Discount Adjusted' values. The console output shows the test results, including the t-statistic, p-value, and confidence interval.

```
1 library(dplyr)
2 library(psych)
3 df <- read.csv("retail_store_sales.csv")
4
5 print("-----6. Paired t-test-----")
6 df$Discount_Adjusted <- df$Total.Spent - runif(nrow(df), min = 0, max = 50)
7 t_testpaired <- t.test(df$Total.Spent, df$Discount_Adjusted, paired = TRUE)
8 print(t_testpaired)
```

Console Output:

```
DElimiter: ,
chr (6): Transaction ID, Customer ID, Category, Item, Payment Method, Location
dbl (3): Price Per Unit, Quantity, Total Spent
lgl (1): Discount Applied
date (1): Transaction Date

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> View(retail_store_sales)
> library(dplyr)
> library(psych)
> df <- read.csv("retail_store_sales.csv")
> print("-----6. Paired t-test-----")
[1] "-----6. Paired t-test-----"
> df$Discount_Adjusted <- df$Total.Spent - runif(nrow(df), min = 0, max = 50)
> t_testpaired <- t.test(df$Total.Spent, df$Discount_Adjusted, paired = TRUE)
> print(t_testpaired)

Paired t-test

data: df$Total.Spent and df$Discount_Adjusted
t = 189.09, df = 11970, p-value < 2.2e-16
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 24.60539 25.12086
sample estimates:
mean difference
 24.86313
```

Environment Pane:

Object	Class	Size
sales_logic	data.frame	1000 obs. of 16 variables
sales_text	data.frame	1000 obs. of 15 variables
selected_cols	data.frame	200 obs. of 3 variables
shopping_behav...	data.frame	3900 obs. of 18 variables
sorted_by_volu...	data.frame	122 obs. of 7 variables
split_matrix	chr	[1:1000, 1:2] "North" "West" "South..."
starts_with_d	data.frame	200 obs. of 3 variables
t_test_one	List	of 10
t_testpaired	List	of 10